Real-Gas Properties from Cubic Equations of State:
A Python Calculator for PVT, Fugacity, Enthalpy and Entropy

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1 What we set out to do

At low pressure a gas obeys $PV_m = RT$ well enough, but at the pressures met in process equipment it does not. We wanted a tool that gives a quick, numerical answer for how far a real pure gas is from ideal behaviour. The program takes a fluid, a temperature and a pressure and returns the compressibility factor, molar volume, fugacity coefficient, fugacity, and the residual and total enthalpy and entropy. It also plots the isotherm and the fugacity curve. Four cubic equations of state (EOS) are available, namely van der Waals (VDW), Redlich–Kwong (RK), Soave–Redlich–Kwong (SRK) and Peng–Robinson (PR), and five fluids: CO_2, CH_4, C_2H_6, N_2 and H_2O. The theory follows Smith et al. [1].

2 Theory used

2.1 One equation for all four models

Each model can be written as

$$P = \frac{RT}{V_m - b} - \frac{a(T)}{V_m^2 + u\,b\,V_m + w\,b^2}, \qquad a = \Omega_a\,\alpha(T_r)\frac{R^2T_c^2}{P_c}, \quad b = \Omega_b\frac{RT_c}{P_c}, \tag{1}$$

and only the numbers in Table 1 change from one model to the next. Here $u = \sigma + \epsilon$ and $w = \sigma\epsilon$. The quantity $\kappa = d\ln\alpha/d\ln T_r$ is needed later for H^R and S^R.

Table 1: Model constants as coded in the program.

Model	u	w	Ω_a	Ω_b	$\alpha(T_r)$	κ
VDW	0	0	27/64	1/8	1	0
RK	1	0	0.42748	0.08664	$T_r^{-1/2}$	$-1/2$
SRK	1	0	0.42748	0.08664	$[1+m(1-\sqrt{T_r})]^2$	$-m\sqrt{T_r}/[1+m(1-\sqrt{T_r})]$
PR	2	-1	0.45724	0.07780	$[1+m(1-\sqrt{T_r})]^2$	$-m\sqrt{T_r}/[1+m(1-\sqrt{T_r})]$

For SRK, $m = 0.480 + 1.574\omega - 0.176\omega^2$. For PR, $m = 0.37464 + 1.54226\omega - 0.26992\omega^2$, and the program uses the longer polynomial $0.379642 + 1.48503\omega - 0.164423\omega^2 + 0.016666\omega^3$ when $\omega > 0.49$.

2.2 From pressure to Z

Substituting $Z = PV_m/RT$, $A = aP/(RT)^2$ and $B = bP/RT$ into Eq. (1) gives a cubic in Z:

$$Z^3 + c_2Z^2 + c_1Z + c_0 = 0, \quad \begin{aligned} c_2 &= -(1 + B - uB), & c_1 &= A + wB^2 - uB - uB^2, \\ c_0 &= -(AB + wB^2 + wB^3). \end{aligned} \tag{2}$$

2.3 Residual properties and fugacity

With $q = A/B$, $\sigma, \epsilon = \frac{1}{2}(u \pm \sqrt{u^2 - 4w})$ and

$$I = \frac{1}{\sigma - \epsilon}\ln\frac{Z + \sigma B}{Z + \epsilon B} \quad (\text{VDW: } I = B/Z),$$

the expressions used are [1]

$$\ln\phi = Z - 1 - \ln(Z - B) - qI, \quad f = \phi P, \quad \frac{H^R}{RT} = Z - 1 + (\kappa - 1)qI, \quad \frac{S^R}{R} = \ln(Z - B) + \kappa qI. \tag{3}$$

The ideal-gas part uses $C_p^{ig} = a + bT + cT^2 + dT^3$ in J/(mol K) and a reference state of 298.15 K and 1 bar:

$$H^{ig} = \int_{T_{ref}}^{T} C_p^{ig}\,dT, \qquad S^{ig} = \int_{T_{ref}}^{T}\frac{C_p^{ig}}{T}\,dT - R\ln\frac{P}{P_{ref}}, \tag{4}$$

so $H = H^{ig} + H^R$ and $S = S^{ig} + S^R$.

3 How the program works

3.1 Solving the cubic

Rather than iterating, the program solves Eq. (2) in closed form. Writing $Z = t - c_2/3$ turns it into $t^3 + pt + q = 0$ with $p = c_1 - c_2^2/3$ and $q = 2c_2^3/27 - c_2 c_1/3 + c_0$. The sign of $\Delta = (q/2)^2 + (p/3)^3$ tells us what to expect:

- $\Delta > 0$: one real root, from the two cube roots of $-q/2 \pm \sqrt{\Delta}$;
- $\Delta \approx 0$ (tolerance 10^{-14}): a repeated root, $t = 2\sqrt[3]{-q/2}$ and $t = -\sqrt[3]{-q/2}$;
- $\Delta < 0$: three real roots, from the trigonometric form

$$t_k = 2\sqrt{-p/3}\,\cos\!\left[\tfrac{1}{3}\arccos\!\left(\tfrac{3q}{2p}\sqrt{-3/p}\right) - \tfrac{2\pi k}{3}\right], \quad k = 0, 1, 2. \tag{5}$$

Roots closer than 10^{-9} are merged, and any root with $Z \le B$ or $Z \le 0$ is dropped because it would mean $V_m \le b$.

3.2 Choosing a phase

If one root survives, that is the answer. If three survive, the smallest is liquid-like and the largest is vapor-like; the middle one has no physical meaning. The program prints both and the ratio f_L/f_V. The root with the smaller fugacity is the stable one. The program does not by itself claim the two phases are in equilibrium, because three roots exist over a whole range of pressures.

3.3 Code layout

The file has a data dictionary `FLUID_DATA` (T_c, P_c, ω and C_p coefficients), a class `CubicModel` for the calculations and a class `PropertyApp` for the window. The methods of `CubicModel` are in Table 2.

Table 2: Methods of `CubicModel`.

Method	Job
`attraction_terms`	$a(T)$, da/dT and κ for the chosen model
`real_cubic_roots`	closed-form real roots of $z^3 + c_2 z^2 + c_1 z + c_0 = 0$
`find_z_roots`	forms A, B, c_2, c_1, c_0, solves, and keeps roots with $Z > B$
`state_properties`	for one root: Z, V_m, $\ln\phi$, ϕ, f, H^R, S^R, H^{ig}, S^{ig}, H, S
`ideal_gas_terms`	H^{ig} and S^{ig} from the C_p polynomial
`pressure_from_volume`	$P(T, V_m)$ from Eq. (1), used for the isotherm

4 The graphical interface

The component, cubic EOS, temperature (K) and pressure (bar) are entered in the system-definition panel. After Calculate Properties is pressed, the property summary cards show Z, molar volume, fugacity and fugacity coefficient. The lower analysis area uses separate tabs for P-V behaviour, fugacity, and enthalpy and entropy; the detailed thermodynamic values and phase-root information are shown in the corresponding analysis view.

5 Test runs

5.1 Single-phase case: CO_2 at 320 K, 50 bar

Here $T_r = 1.052$ and $P_r = 0.678$, and every model returns one root.

Table 3: CO_2 at 320 K and 50 bar, as printed by the program.

Model	Z	V_m [cm^3/mol]	ϕ	f [bar]	H^R [kJ/mol]	S^R [J/mol K]	H [kJ/mol]	S [J/mol K]
VDW	0.7854	417.9	0.8239	41.19	−1.446	−2.908	−0.623	−32.77
RK	0.7689	409.2	0.8079	40.40	−1.893	−4.143	−1.071	−34.01
SRK	0.7761	413.0	0.8124	40.62	−2.142	−4.967	−1.319	−34.83
PR	0.7541	401.3	0.7935	39.68	−2.210	−4.983	−1.387	−34.85

All four give $Z < 1$ and $\phi < 1$, which means attraction between molecules outweighs repulsion here and the fugacity (≈ 40 bar) is below the 50 bar pressure. PR gives the densest fluid. VDW has the smallest $|H^R|$; its $\alpha = 1$ makes $\kappa = 0$, so the temperature dependence of a contributes nothing to H^R. The totals include $H^{ig} = 0.823$ kJ/mol and $S^{ig} = -29.86$ J/(mol K) for the PR case.

5.2 Three roots: CH_4 at 160 K, 10 bar (PR)

The program finds $Z = 0.0342$, 0.0895 and 0.8562 (Fig. 2). The liquid-like root gives $\phi = 1.253$ and the vapor-like root gives $\phi = 0.872$, so $f_L/f_V = 1.44$ and the vapor is stable at this pressure. The isotherm plot hides the part of the loop where the model pressure is negative, since the code only plots positive pressures.

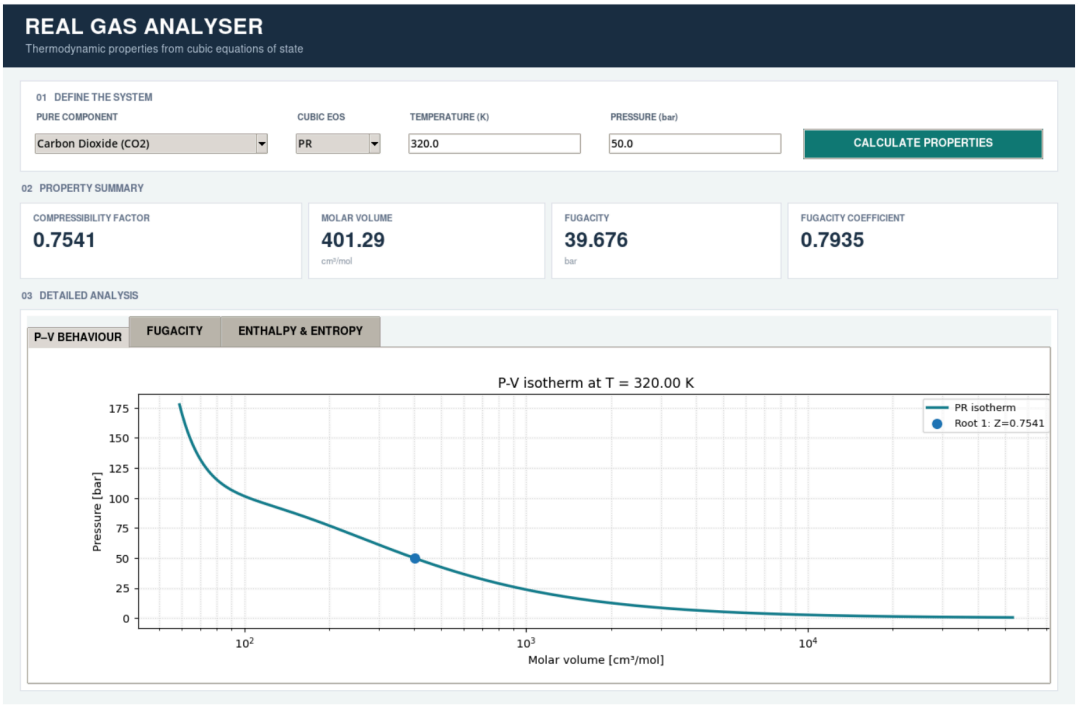


Figure 1: Dashboard interface for CO2, PR, T = 320 K and P = 50 bar, showing the property summary and P-V analysis tab.

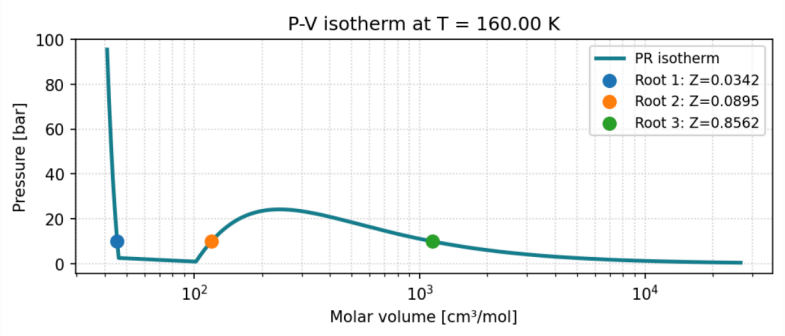


Figure 2: CH_4 isotherm at 160 K (PR) with the three roots at 10 bar.

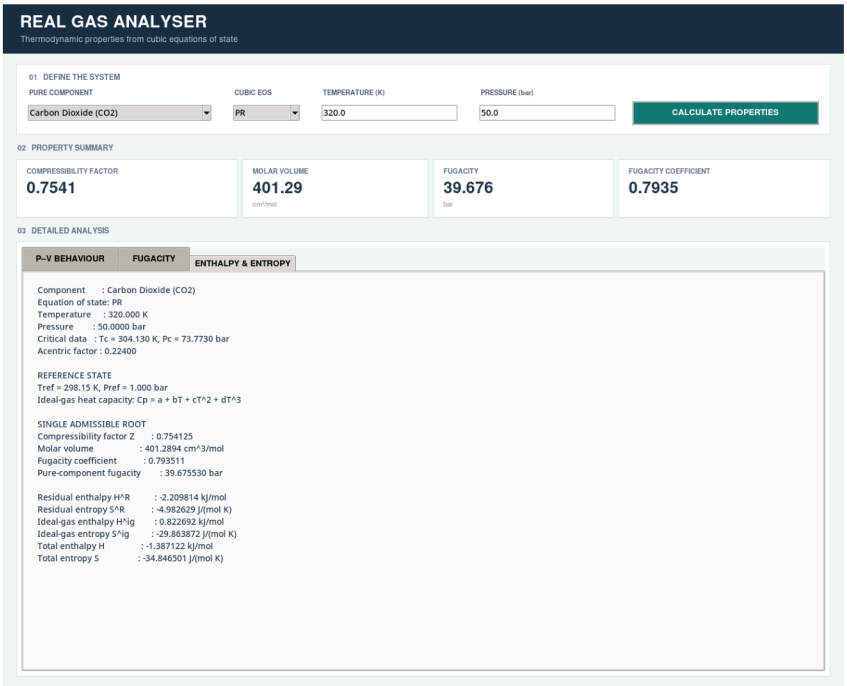


Figure 3: Dashboard showing the calculated thermodynamic results for CO2 using PR EOS at 320 K and 50 bar.

5.3 Saturation pressure from $f_L = f_V$

For a few fluids we located the pressure where the two outer roots have equal fugacity, using bisection on $\ln\phi_L - \ln\phi_V$ with the program's own `find_z_roots` and `state_properties`. This search was a separate script and is not part of the submitted program. At equal T and P the ideal-gas terms cancel, so $\Delta H^{vap} = H_V^R - H_L^R$. As a check, the Clapeyron relation $\Delta S^{vap} = \Delta H^{vap}/T$ holds for CH_4 (PR): $S_V^R - S_L^R = 37.06$ and $\Delta H^{vap}/T = 37.06$ J/(mol K). PR gives a slightly lower P^{sat} and smaller Z_L, V_L than SRK in every row. The PR value for CO_2 at 280 K is close to the experimental figure of about 41.6 bar.

Table 4: Saturation states found from equal fugacity.

Fluid	Model	T [K]	P^{sat} [bar]	Z_L	Z_V	V_L [cm^3/mol]	V_V [cm^3/mol]	ΔH^{vap} [kJ/mol]
CH_4	PR	160	16.05	0.0542	0.7474	44.9	619.3	5.930
CH_4	SRK	160	16.15	0.0618	0.7598	50.9	625.8	5.988
CO_2	PR	280	41.60	0.0923	0.6412	51.7	358.8	9.147
CO_2	SRK	280	41.99	0.1054	0.6578	58.4	364.7	9.110
C_2H_6	PR	250	13.05	0.0402	0.7948	64.0	1266.1	11.227
C_2H_6	SRK	250	13.17	0.0460	0.8042	72.6	1269.0	11.309
N_2	PR	100	7.805	0.0348	0.8148	37.0	868.0	4.503
N_2	SRK	100	7.850	0.0396	0.8237	42.0	872.5	4.556

6 Where the method falls short

- **Water.** At 373.15 K, PR gives $P^{sat} = 0.965$ bar against the steam-table 1.013 bar (about 4.8% low); SRK gives 0.928 bar. The $\alpha(T_r)$ term uses only T_r and ω, so it cannot describe hydrogen bonding.
- **Near the critical point** cubic EOS are least reliable and overpredict Z_c.
- **Input data.** Critical constants and C_p coefficients are approximate values typed into the program, and the C_p polynomials are meant for temperatures above roughly 298 K, so total H and S at low T are extrapolated. Only pure fluids are handled.

7 Summary

The program returns Z, Vm, phi, f, HR, SR, H and S for five pure fluids using VDW, RK, SRK or PR, and plots the isotherm and fugacity curve. The cubic is solved in closed form; phase behaviour is assessed using fugacity ratios. For CO2 at 320 K and 50 bar, the models give Z from 0.754 to 0.785 and f from 39.7 to 41.2 bar. Equal-fugacity saturation pressures satisfy the Clapeyron check; the main weakness is polar water.

References

[1] J. M. Smith, H. C. Van Ness, M. M. Abbott and M. T. Swihart, Introduction to Chemical Engineering Thermodynamics, 8th ed., McGraw-Hill, 2018.